

Question 1 : Describe different types of data sources used in ETL with suitable examples.

Answer:

In ETL (Extract, Transform, Load), data sources are the locations or systems from which data is collected for processing and analysis. Different types of data sources can be used depending on the business requirement.

1. Relational Databases

Relational databases store data in tables consisting of rows and columns. They are commonly used in organizations for storing structured business data.

Examples:

- MySQL
- PostgreSQL
- Oracle
- Microsoft SQL Server

Example: A company's customer and sales data stored in a MySQL database.

2. Flat Files

Flat files contain data in a simple file-based format.

Examples:

- CSV
- TSV
- Text files

Example: A monthly sales report stored in a CSV file.

3. Excel Files

Excel spreadsheets are commonly used by businesses to store and exchange structured data.

Example: An employee attendance report stored in an `.xlsx` file.

4. APIs

APIs allow applications to communicate with external systems and retrieve data.

Examples:

- Weather APIs
- Payment APIs
- Social media APIs
- E-commerce APIs

Example: An ETL process extracting current exchange-rate information from a financial API.

5. NoSQL Databases

NoSQL databases are useful for storing semi-structured or unstructured data.

Examples:

- MongoDB
- Cassandra
- Redis

Example: Customer activity data stored as JSON documents in MongoDB.

6. Cloud Storage

Cloud platforms can store large amounts of files and data that can be extracted by ETL tools.

Examples:

- Amazon S3
- Google Cloud Storage
- Azure Blob Storage

Example: An organization storing daily transaction files in Amazon S3.

7. Web and Application Data

Data can also come from websites, web applications, logs, and other digital systems.

Example: Website clickstream or server log data collected for customer behavior analysis.

Therefore, ETL developers may work with structured, semi-structured, and unstructured data sources depending on the type of information required.

Question 2 : What is data extraction? Explain its role in the ETL pipeline.

Answer:

Data extraction is the process of collecting or retrieving data from one or more source systems so that it can be processed and used in an ETL pipeline.

ETL stands for:

E – Extract

T – Transform

L – Load

Role of Data Extraction in ETL

Extraction is the first stage of the ETL process. Its main purpose is to obtain the required data from source systems without losing important information.

The extraction process generally involves:

1. Identifying the required source systems.
2. Connecting to those systems.
3. Selecting the required data.
4. Reading or extracting the data.
5. Sending the extracted data to the transformation stage.

Example:

Suppose a retail company stores sales information in a database:

```
ProductID | ProductName | Quantity | Price
```

The ETL process may extract this sales data from the database, transform it by calculating revenue or cleaning incorrect values, and then load the processed data into a data warehouse.

Importance of Data Extraction

Data extraction is important because:

- It provides the raw data required for analysis.
- It brings data from multiple sources into the ETL pipeline.
- It allows organizations to combine data from different systems.
- It forms the foundation for the transformation and loading stages.

In simple terms, data extraction collects the required raw data from source systems so that it can be cleaned, transformed, and loaded into the target system.

Question 3 : Explain the difference between CSV and Excel in terms of extraction and ETL usage.

Answer:

CSV and Excel are both commonly used as data sources in ETL, but they differ in structure and features. CSV files are simple, lightweight, and contain data in rows and columns, making them easy and fast to extract and process in ETL tools. They generally do not support features such as formulas, formatting, or multiple worksheets.

Excel files, on the other hand, support multiple sheets, formulas, formatting, and more complex data structures. Because of these features, extracting data from Excel can require additional processing compared to CSV.

CSV is generally preferred for simple and efficient ETL data extraction, while Excel is useful when the data contains multiple sheets or spreadsheet-specific features.

Question 4 : Explain the steps involved in extracting data from a relational database.

Answer:

Data extraction from a relational database involves connecting to the database, identifying the required tables and columns, and using SQL queries to retrieve the necessary data. The ETL developer may perform either a full extraction, where all required records are extracted, or an incremental extraction, where only new or updated records are extracted. The extracted data is then validated and passed to the transformation stage.

In summary, the main steps are connecting to the database, selecting the required data, executing the extraction query, validating the extracted data, and sending it to the next stage of the ETL pipeline.

Question 5 : Explain three common challenges faced during data extraction.

Answer:

Three common challenges in data extraction are large data volume, inconsistent data, and connectivity or access problems. Large amounts of data can require significant memory, processing time, and network resources. Data from different sources may also have different formats, data types, or missing values, which makes extraction and processing more difficult. In addition, network failures, incorrect credentials, database downtime, or permission issues can interrupt the extraction process.

In summary, successful data extraction requires proper handling of large datasets, data quality and consistency, and reliable access to the source systems.

Question 6 : What are APIs? Explain how APIs help in real-time data extraction.

Answer:

API (Application Programming Interface) is a way that allows different software applications to communicate and exchange data. In ETL, APIs help retrieve data directly from external applications or services without requiring manual file downloads. This makes it possible to collect frequently updated or near-real-time data from sources such as weather services, financial systems, e-commerce platforms, and other applications.

In summary, APIs provide an automated way to access current data from external systems, making them useful for real-time or near-real-time data extraction in ETL pipelines.

Question 7 : Why are databases preferred for enterprise-level data extraction?

Answer:

Databases are preferred for enterprise-level data extraction because they can handle large volumes of structured data and provide efficient ways to retrieve the required information. They support SQL queries, security controls, data integrity, and reliable access, which are important for enterprise systems. Databases also make it easier to extract only the required records and support techniques such as incremental extraction.

In summary, databases are preferred because they provide scalability, efficient querying, security, data integrity, and reliable management of large amounts of enterprise data.

Question 8 : What steps should an ETL developer take when extracting data from large CSV files (1GB+)?

Answer:

When extracting data from large CSV files of 1GB or more, an ETL developer should avoid loading the entire file into memory at once. The file should be processed in smaller **chunks or batches** to reduce memory usage. The developer should also select only the required columns, validate the data, handle missing or invalid records, and maintain proper error logging. Performance should be monitored to ensure efficient processing.

In summary, for large CSV files, the main steps are chunk-based processing, selecting required data, validating records, handling errors, and monitoring performance to make the extraction process efficient and reliable.